

Software Test Plan (STP) – Collaborative Study Material & Quiz Platform

Project: Collaborative Study Material & Quiz Platform

Version: 1.0

Authors: Group 5 (Bharat Shandilya, Sandeep H, Samyak Sanklecha, Shreyas B Patil)

Date: 17-09-2026

1. Introduction

Purpose: This document defines the test plan for the Collaborative Study Material & Quiz Platform v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing the platform before demonstration/evaluation.

Scope: Testing covers the core modules of the platform — study material sharing, topic-based question bank, quiz creation and attempting, performance analysis and progress tracking, feedback/ratings, and administration. Since the system is demonstrated on localhost, testing of public deployment, cloud hosting, and third-party payment/authentication integrations is out of scope (none are used in this version).

References: SRS – Collaborative Study Material & Quiz Platform v1.0, Project Proposal – Collaborative Study Material & Quiz Platform, Course guidelines (UE24CS341A).

Definitions: MERN (MongoDB, Express.js, React, Node.js), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing).

2. Test Items

- Authentication module (registration/login)
- Study Material Sharing module
- Question Bank module
- Quiz Creation & Attempting module
- Performance Analysis / Dashboard module
- Feedback & Ratings module
- Admin module (user, content, subject/topic management)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- REQ-1 to REQ-3: Study material upload, browse/search/filter, error handling
- REQ-4 to REQ-6: Add question, search/filter questions, centralized question storage
- REQ-7 to REQ-9: Quiz creation, attempting, auto-evaluation of objective questions

- REQ-10 to REQ-12: Recording attempts, score/topic-wise performance, performance-over-time view
- REQ-13 to REQ-14: Ratings/feedback submission and average rating display
- REQ-15 to REQ-17: Admin user management, content moderation, subject/topic management
- Non-functional: Response time (~2–3s for search/load, ~2s for score calc, ~3s for dashboard), support for 20–30 concurrent users, password hashing, role-based access

4. Features Not to be Tested

- MongoDB internal engine behavior (assumed reliable, third-party)
- Underlying MERN framework/library internals (React, Express, Node runtime)
- Public deployment / cloud hosting (not implemented in this version — localhost only)
- Payment and internationalisation features (explicitly out of scope per SRS)

5. Test Approach / Strategy

Levels:

- Unit tests (module-level — e.g., individual API routes/controllers)
- Integration tests (React frontend ↔ Node/Express backend ↔ MongoDB)
- System tests (end-to-end flows across modules, e.g., upload material → search → use in quiz)
- Acceptance tests (UAT with faculty/evaluator during demo)

Types:

- Functional testing (all core features — material sharing, question bank, quizzes, dashboard, feedback, admin)
- Regression testing (after each module integration)
- Performance testing (response time under the 20–30 concurrent user target)
- Usability testing (navigation clarity, error messages, dashboard readability)

Entry Criteria: Stable local build available, MongoDB running locally, test data (dummy users/materials/questions) available, test environment (localhost) ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all high-priority functional requirements (REQ-1 to REQ-9) satisfied.

5.1 Security Validation

- Verify passwords are stored in hashed form, not plain text
- Verify unauthenticated users cannot upload material, add questions, attempt quizzes, or give feedback
 - Verify admin-only actions (manage users/content/topics) are restricted to the admin role
- Basic input validation checks on file uploads (unsupported file type handling) and form

fields

6. Test Environment

Hardware: Standard laptops used by the team (no special hardware required). **Software:** React frontend, Node/Express backend, MongoDB (local instance), modern browser (Chrome). **Tools:** Postman (API testing), Jira (task/defect tracking), GitHub (version control), browser DevTools (frontend debugging). **Test Data:** Dummy student/contributor/admin accounts, sample study materials (PDFs/notes), sample questions across subjects/topics/difficulty levels.

7. Test Schedule

Milestones:

- Test case design: 27-09-2026
- Environment setup: 27-09-2026
- Test execution start: 27-09-2026
- Test execution end: 30-09-2026
- UAT / Faculty demo: 30-09-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual)
- Test Data
- Test Execution Logs
- Defect Reports (via Jira)
- Test Summary Report

9. Roles and Responsibilities

		Role Name Responsibility
QA Lead / Coordinator	Group 5 (shared)	Prepare plan, coordinate execution
Materials & Feedback	Module Owner Samyak Sanklecha	Test material sharing and ratings/feedback module
Question Bank	Module Owner Sandeep H	Test question bank add/search/filter
Quiz	Module Owner Bharat Shandilya	Test quiz creation, attempting, auto-evaluation
Performance	Module Owner Shreyas B Patil	Test performance dashboard and progress tracking

Admin Module Group 5 (shared) Test admin user/content/topic management
Faculty / Evaluator Dr. Sudeepa Roy Dey Review and approve UAT/demo results

10. Risks and Mitigation

Risk Mitigation

Delay in module integration before demo	Integrate modules early and test incrementally rather than all at once
Local environment inconsistencies across team members' laptops	Maintain a shared setup guide / consistent Node & MongoDB versions
Limited time within semester schedule	Prioritize testing on high-priority (REQ-1–9) features first

11. Assumptions & Dependencies

- The system will be tested on localhost only, matching the demonstration setup
- Test data (dummy accounts, materials, questions) will be prepared before execution
- Open-source MERN components remain freely available and stable during testing

12. Suspension & Resumption Criteria

Suspend testing if:

- Local environment (MongoDB/backend) unavailable for an extended period
- Build is too unstable (blocks a large portion of planned test cases)

Resume testing if:

- Blocking defects are resolved
- Local environment is stabilized

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.

Example:

- REQ-1 (Upload material) → TC-Mat-01, TC-Mat-02

- REQ-7 (Create quiz) → TC-Quiz-01, TC-Quiz-02
- REQ-9 (Auto-evaluation) → TC-Quiz-03
- REQ-11 (Performance display) → TC-Perf-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect count by module
- Requirement coverage (REQ-1 to REQ-17)

Reports:

- Execution status updates during testing phase
- Final Test Summary Report (before UAT/demo)

15. Approvals

Role Name Signature / Date Team Member Sandeep H Sandeep H/17-09-2026

Team Member Bharat Shandilya Bharat Shandilya/17-09-2026

Team Member

Samyak Sanklecha Samyak Sanklecha/17-09-2026

Team Member

Faculty (Dr. Sudeepa Roy Dey)